

SHETH LUJ AND SIR MV COLLEGE
Subject: Data Analysis with SAS / SPSS / R

M2 Practical No: 9

Aim: Conducting Chi-square tests using `chisq.test()` (R)

Code:

```
data <- read.csv("meteorite-landings.csv", nrow = 5)

median_revenue <- median(data$mass, na.rm = TRUE)
median_net_income <- median(data$year, na.rm = TRUE)

data$Revenue_Cat <- ifelse(data$mass >= median_revenue, "High", "Low")
data$Net_Income_Cat <- ifelse(data$year >= median_net_income, "High", "Low")

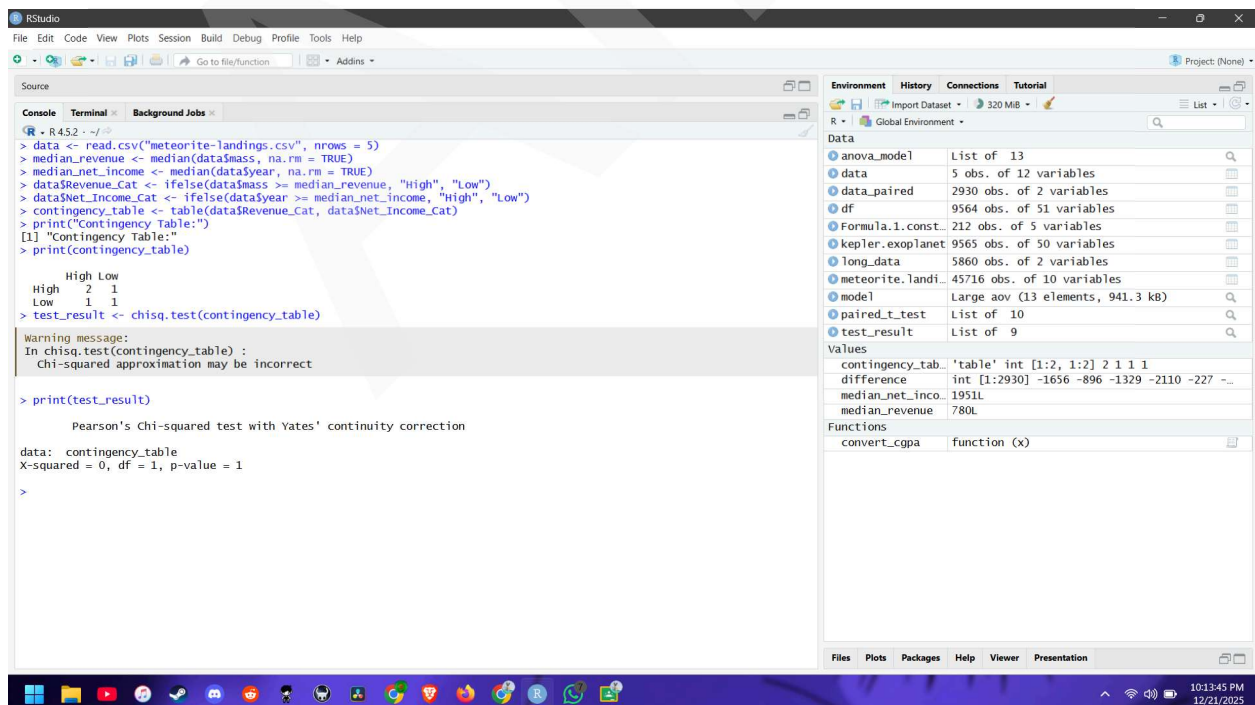
contingency_table <- table(data$Revenue_Cat, data$Net_Income_Cat)

print("Contingency Table:")
print(contingency_table)

test_result <- chisq.test(contingency_table)

print(test_result)
```

Output:



```
R - R4.5.2 ~ /
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function Addins
Source
Console Terminal Background Jobs
> data <- read.csv("meteorite-landings.csv", nrow = 5)
> median_revenue <- median(data$mass, na.rm = TRUE)
> median_net_income <- median(data$year, na.rm = TRUE)
> data$Revenue_Cat <- ifelse(data$mass >= median_revenue, "High", "Low")
> data$Net_Income_Cat <- ifelse(data$year >= median_net_income, "High", "Low")
> contingency_table <- table(data$Revenue_Cat, data$Net_Income_Cat)
> print("Contingency Table:")
[1] "Contingency Table:"
> print(contingency_table)
      High Low
High    2   1
Low     1   1
> test_result <- chisq.test(contingency_table)
Warning message:
In chisq.test(contingency_table) :
  Chi-squared approximation may be incorrect
> print(test_result)
Pearson's Chi-squared test with Yates' continuity correction

data: contingency_table
X-squared = 0, df = 1, p-value = 1
>
```

Environment History Connections Tutorial

R • Global Environment • 320 MiB •

Data

- anova_model List of 13
- data 5 obs. of 12 variables
- data_paired 2930 obs. of 2 variables
- df 9564 obs. of 51 variables
- Formula.1.const 212 obs. of 5 variables
- kepler.exoplanet 9565 obs. of 50 variables
- long_data 5860 obs. of 2 variables
- meteorite.landi_ 45716 obs. of 10 variables
- model Large aov (13 elements, 941.3 kB)
- paired_t_test List of 10
- test_result List of 9

Values

contingency_tab	'table' int [1:2, 1:2] 2 1 1 1
difference	int [1:2930] -1656 -896 -1329 -2110 -227 ...
median_net_inco	1951L
median_revenue	780L

Functions

convert_cgpa	function (x)
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